

Connectivity and Interoperability in the Apple Ecosystem (2017–2020)

Introduction

From 2017 through 2020, Apple's hardware ecosystem underwent significant transitions in connectivity and interoperability. This period saw the standardization of Thunderbolt 3 across Macs, the maturation of wireless protocols like 802.11ac Wi-Fi and Bluetooth 5.0, and the expansion of Apple's Continuity features linking Macs with iPhones, iPads, and peripherals. This paper provides a technical analysis of these developments, focusing on how two example machines – a 2017 iMac 5K (27-inch) and a 2019 16-inch MacBook Pro – leverage Apple's connectivity advancements. We compare wired interfaces (Thunderbolt versions and USB), wireless capabilities (Wi-Fi and Bluetooth standards), peer-to-peer sharing mechanisms (notably AirDrop over Apple Wireless Direct Link), and ecosystem integration features (Continuity, Sidecar, etc.). We also examine how these Macs fit into multi-device workflows, their roles in OS deployment and security (e.g. the T2 security chip), peripheral compatibility (external displays, eGPUs, etc.), and upgrade paths. Through detailed specifications and comparative tables, we highlight both the commonalities and key differences that define the connectivity performance of the 2017 iMac and 2019 MacBook Pro within Apple's broader ecosystem.

Wired Connectivity: Thunderbolt and USB

Thunderbolt Evolution (TB1 to TB3): Apple has consistently pushed high-bandwidth interfaces, beginning with Thunderbolt 1 in 2011 (10 Gb/s per channel) and Thunderbolt 2 in 2013 (20 Gb/s). Thunderbolt 3, introduced in late 2015, doubled the bandwidth to 40 Gb/s and crucially adopted the USB-C connector, merging data, video, and power into one port ¹ ². A single Thunderbolt 3 (TB3) port carries a PCI Express (PCIe) data bus and dual DisplayPort 1.2 streams, enabling *simultaneous* high-speed peripheral and display connections ³. For example, one TB3 link supports **two** 4K monitors or one 5K monitor at 60 Hz from a single port ⁴. The 40 Gb/s bandwidth is also enough to handle massive data transfers while driving displays, making TB3 a “computer port nirvana” that fulfills the promise of USB-C for one-cable docking, external GPUs, and more ⁵ ⁶. TB3 introduced support for USB 3.1 Gen2 (10 Gb/s) natively over the USB-C plug, meaning the same port can connect billions of USB devices at up to 10 Gb/s ². Furthermore, TB3 integrates **100 W power delivery** for charging and can supply up to 15 W to bus-powered accessories ⁷. Notably, Thunderbolt remains backward-compatible: older Thunderbolt 1/2 devices (Mini DisplayPort connector) work via a USB-C adapter without performance loss ⁸. By 2017, Apple had fully embraced Thunderbolt 3 on Macs – including the iMac and MacBook Pro in this study – standardizing on the USB-C port for all high-speed wired connectivity.

2017 iMac 5K Wired Interfaces: The Mid-2017 iMac 27” was Apple's first 5K iMac to feature Thunderbolt 3/USB-C ports (replacing the Thunderbolt 2 ports of the 2015 model). It provides *two* TB3 (USB-C) ports on the rear, each supporting the full 40 Gb/s throughput, DisplayPort alt-mode, USB 3.1 Gen2 (10 Gb/s), and daisy-chaining of up to 6 devices ⁹ ⁶. In addition, the iMac caters to legacy peripherals with four USB-A 3.0 ports (5 Gb/s each), an SDXC card slot, and a Gigabit Ethernet RJ-45 port ¹⁰ ¹¹. The TB3 ports on iMac can

natively drive one external 5K display or dual 4K displays at 60 Hz alongside the built-in 5K screen ¹². For example, an iMac 5K (with a discrete Radeon Pro 5xx GPU) can run its internal 5120×2880 display plus one additional 5K monitor via TB3, or two 4K UHD monitors, all at 60 Hz ¹². These versatile ports also support adapters for HDMI, DVI, VGA, or Thunderbolt 2 devices ¹³ ¹⁴. The inclusion of dedicated USB-A and SD slots reflects the iMac's role as a workstation hub, ensuring compatibility with common peripherals (USB drives, SD cards, etc.) without dongles.

2019 MacBook Pro 16" Wired Interfaces: The 16-inch MacBook Pro (2019) takes a more minimalist, forward-looking approach: it includes *four* Thunderbolt 3 (USB-C) ports (two on each side), and a 3.5 mm audio jack – and nothing else in terms of physical ports ¹⁵ ¹⁶. All four USB-C ports are fully TB3-enabled with charging capability, allowing the MacBook to be charged from any port via USB-C power delivery ¹⁷. Each port supports the same multi-protocol functions (40 Gb/s Thunderbolt, 10 Gb/s USB 3.1, DisplayPort video) and can drive external displays or high-speed storage. Thanks to a high-end AMD Radeon Pro GPU (and two TB3 controllers to distribute bandwidth), the 2019 MBP 16" can drive up to **two 6K displays** (6016×3384, such as Apple's Pro Display XDR) or as many as four 4K monitors at 60 Hz simultaneously ¹⁸ – a notable increase over the 2017 iMac's external display limit. However, this expandability comes at the cost of legacy ports: the MBP offers no built-in USB-A, SD card, or Ethernet. Professional users must use adapters or docks (e.g. USB-C to USB-A or SD readers) for those needs ¹⁶. This design reflects Apple's aggressive adoption of USB-C and reliance on wireless alternatives, but it drew some criticism since *"the port selection has not changed – you're still living the dongle life"* for things like USB sticks or SD cards ¹⁶. In summary, the MacBook Pro's wired connectivity is entirely Thunderbolt/USB-C centric, whereas the iMac provides a broader *mix* of new and legacy ports – an important distinction in connectivity use-cases.

Use of Thunderbolt for Peripherals and Networking: Both the iMac and MacBook Pro exploit Thunderbolt 3 for high-performance peripherals. External Thunderbolt RAID drives or NVMe SSD enclosures can achieve transfer speeds in the 2–3 GB/s range, leveraging the 40 Gb/s (~5 GB/s) raw bandwidth – far beyond the 5 Gb/s of the iMac's internal Fusion Drive HDD portion. Professionals frequently use TB3 docks or hubs to connect multiple devices: a single TB3 port can daisy-chain a 5K display, external GPU (eGPU), audio interface, and 10 Gb Ethernet adapter together. In fact, Apple officially supported eGPUs starting in macOS High Sierra 10.13.4 (2018) for TB3 Macs. Both the 2017 iMac and 2019 MBP can connect to an external AMD Radeon eGPU chassis to accelerate graphics for tasks like 3D rendering or driving VR headsets. Apple's Metal API and apps like Final Cut Pro utilize eGPU horsepower when available. Another capability of Thunderbolt 3 is **peer-to-peer networking**: two Macs can be connected directly via a TB3 cable for a 10 Gigabit Ethernet (10 GbE) network link ¹⁹. This is useful for fast data migration or clustering machines – effectively treating the TB cable as a virtual 10 Gb network interface. Both example Macs support this "Thunderbolt Networking" feature (the MacBook Pro even has TB3 on both sides, offering flexibility in cabling). Finally, Thunderbolt 3's ability to carry DisplayPort makes it the default way to connect external monitors on these Macs. The 2017 iMac's TB3 ports double as outputs for second displays, and the MacBook Pro uses its TB3/USB-C ports for all video output (with USB-C-to-DisplayPort/HDMI dongles as needed). In sum, Thunderbolt 3 in the 2017–2020 Apple lineup represents a unification of connectivity – combining PCIe, USB, display, and power – that enabled new classes of peripherals (eGPUs, one-cable docks) and simplified multi-monitor setups ⁵ ⁶.

Wireless Networking: Wi-Fi and Beyond

802.11ac Wi-Fi (Wi-Fi 5): Both the 2017 iMac and 2019 MacBook Pro are equipped with 802.11ac Wi-Fi, known as "Wi-Fi 5." Apple adopted 802.11ac early (initially on MacBook Air/Pro in 2013), and by 2017 all

Macs featured it. The iMac (2017) supports 802.11ac in a dual-band 2.4/5 GHz implementation with MIMO (multiple antennas) ²⁰. The MacBook Pro (2019) likewise has 802.11ac with backward compatibility to 802.11a/b/g/n networks ²¹. In practice, 802.11ac on these Macs can achieve theoretical link speeds up to about **1.3 Gb/s** (if a 3×3 MIMO configuration at 80 MHz channel width is supported). Real-world throughput tends to be hundreds of megabits per second, which was a huge jump over the previous 802.11n standard. This “Gigabit-class” Wi-Fi was critical for Apple’s ecosystem features like AirDrop and iCloud syncing to feel fast. The iMac, being a desktop, also includes a Gigabit Ethernet port – often yielding lower latency and consistent speeds for local transfers or network storage. The MacBook Pro has no built-in Ethernet, but users can utilize a Thunderbolt or USB-C Gigabit Ethernet adapter if needed. Still, Apple clearly expects most MacBook Pro users to rely on Wi-Fi (or Thunderbolt dock connections) for networking. Both example machines support personal hotspot tethering to an iPhone via Wi-Fi or Thunderbolt, providing flexibility for internet access on the go.

During 2017–2020, the newer **802.11ax Wi-Fi (Wi-Fi 6)** standard emerged, promising up to 40% higher throughput, lower latency, and better performance in dense environments. Notably, Apple introduced Wi-Fi 6 in the iPhone 11 (2019) and iPad Pro (2020), but **no Intel-based Mac through 2020 included Wi-Fi 6** support ²². The first Macs to get 802.11ax were the Apple Silicon M1 models released in late 2020 ²². Thus, our iMac and MBP (being Intel-based) are limited to Wi-Fi 5. In practical terms, this means their wireless speed and range are very similar to one another. The iMac’s larger chassis may confer some RF advantage (potentially housing higher-gain antennas), whereas the MacBook Pro’s all-metal body and portability mean it might have slightly less Wi-Fi range. Nonetheless, both can comfortably exceed 500 Mb/s under ideal conditions on 5 GHz 802.11ac networks – more than sufficient for tasks like high-bitrate video streaming or fast Time Machine backups to a network drive. It’s worth noting that Apple’s implementation of 802.11ac also leverages **beamforming** and other features of the standard to improve reliability.

Apple Wireless Direct Link (AWDL): In addition to infrastructure Wi-Fi, Apple devices in this era make heavy use of a proprietary peer-to-peer Wi-Fi protocol called **AWDL**. AWDL is a *mesh-capable* ad-hoc protocol built on top of Wi-Fi that allows devices to discover and communicate with each other directly (without an access point) ²³ ²⁴. It underpins features like AirDrop, AirPlay peer-to-peer, and GameKit networking. AWDL works by **channel hopping**: even while a Mac or iPhone is connected to a normal Wi-Fi network, the wireless chipset can rapidly switch to a special “social” channel to send discovery broadcasts and then jump to a designated channel for direct data transfer between peers ²⁴. This happens in the background, fast enough that the user’s main Wi-Fi connectivity is not disrupted beyond a minor performance impact ²⁵. Both the 2017 iMac and 2019 MBP support AWDL through their Broadcom (or similar) Wi-Fi chipsets. For instance, if one wants to AirDrop a large file from the MacBook Pro to the iMac, the two will use Bluetooth Low Energy to locate each other, then seamlessly create an AWDL link over Wi-Fi for the file transfer (more on AirDrop below). The advantage of AWDL is that it doesn’t require the devices to be on the same Wi-Fi network – they can form a private, high-speed link anywhere, even in a field with no router ²³ ²⁶. Apple’s use of AWDL in this period was pioneering, although it is a proprietary protocol. (It was later revealed that regulatory pressure in some regions might push Apple to adopt the open Wi-Fi Aware standard instead of AWDL in the future, to ensure interoperability ²⁷ ²⁸.)

Bluetooth Classic and 5.0: Alongside Wi-Fi, Apple continued to update Bluetooth capabilities in Macs for short-range connectivity. The 2017 iMac comes with **Bluetooth 4.2**, an update over BT 4.0 that offers improved speed (Bluetooth 4.2 supports up to ~1 Mb/s for Low Energy data) and privacy features ²⁹. Bluetooth is crucial for connecting input devices (the iMac’s included Magic Keyboard and Magic Mouse 2 use BT), as well as Continuity features like Handoff and Universal Clipboard (which use BLE advertisements).

The 2019 MacBook Pro features **Bluetooth 5.0**, which doubled the Low Energy PHY rate to 2 Mb/s and optionally increases range up to 4× over BLE 4.x (at the cost of speed) ³⁰ ³¹ . In practical terms, BT 5.0 on the MBP gives more robust connections for devices like AirPods or Bluetooth speakers and can maintain continuity connections at greater distances from an iPhone. Apple's macOS and iOS use BLE (Bluetooth Low Energy) extensively for device discovery – e.g., the MacBook Pro can detect an Apple Watch nearby to allow auto-unlock, or discover that an iPhone is ringing to relay a call. Both BT 4.2 and 5.0 are backward-compatible, so all common BT peripherals (keyboards, mice, headphones) work with both Macs. However, the MBP's BT 5.0 potentially offers better performance for BLE data transfers involved in AirDrop and Handoff. For example, when using AirDrop, initial negotiation occurs over BLE – BT 5.0's higher data rate could slightly speed up the handshake and make discovery more responsive (though the bulk file transfer uses Wi-Fi). It's also notable that by having Bluetooth, both Macs support **Continuity** features like Universal Clipboard or Handoff as long as they are within BLE range (on the order of 10 m).

In summary, the wireless connectivity on the 2017 iMac and 2019 MBP is quite similar in generation (Wi-Fi 5, BLE-based BT), ensuring both can fully participate in Apple's wireless ecosystem. The MacBook Pro's Bluetooth 5.0 is a step up from the iMac's 4.2, aligning with Apple's gradual rollout (Apple added BT 5.0 starting in some 2018 Macs and all 2019 portable Macs ³² ³³). On Wi-Fi, any performance differences would stem more from antenna design than protocol, since both are using the same 802.11ac standard. For critical tasks, both machines could fall back to wired Gigabit Ethernet (the iMac directly, the MBP via an adapter). By 2020, users anticipating Wi-Fi 6 had to wait for the Apple Silicon Macs – an important note for “future-proofing,” as Apple did not refresh Intel Macs with Wi-Fi 6 at all ²² . Thus, in an enterprise or home setup with a new Wi-Fi 6 router, these 2017–2019 Macs would operate at 802.11ac speeds (albeit still very fast).

Peer-to-Peer Sharing: AirDrop and AirPlay

One of the hallmark interoperability features in Apple's ecosystem is **AirDrop**, introduced on Mac in 2011 and on iOS in 2013. AirDrop allows quick, ad-hoc file transfer between nearby Apple devices (Mac, iPhone, iPad) using a combination of Bluetooth and Wi-Fi. By the time of macOS High Sierra / iOS 11 (circa 2017), AirDrop had evolved into a seamless cross-platform protocol leveraging Apple's AWDL technology. Here is how it works technically:

- **Discovery:** When AirDrop is enabled, a device advertises itself via Bluetooth Low Energy. For example, the 2019 MacBook Pro will periodically send BLE advertisement packets that include information (encrypted) about the device. Nearby Apple devices pick up these adverts. Bluetooth provides a power-efficient way to discover devices in proximity (within ~30 ft). On iOS, starting iOS 13, an ultra-wideband chip (U1) can augment this discovery with directional awareness (though Macs in 2019 didn't have U1, only iPhones did). The 2017 iMac can similarly see nearby AirDrop-enabled devices via BLE, despite being a desktop – this allows an iMac to appear in the AirDrop share sheet on an iPhone, for instance.
- **Connection setup:** Once two devices mutually discover each other via Bluetooth and determine (through exchanged info) that they both support AirDrop and that the user permits receiving (contacts or everyone), they initiate a direct Wi-Fi link. This is where AWDL comes in. The devices coordinate to create a peer-to-peer Wi-Fi session on a channel that both set aside for AirDrop ²⁴ . They switch their Wi-Fi radios to that channel, establish a link-local IPv6 network between them, and then switch back and forth as needed (if they also remain on an infrastructure network) ³⁴ ²⁴ . They

negotiate encryption keys using a built-in **TLS** handshake – AirDrop communication is secured by TLS over this direct Wi-Fi link ³⁵ . Notably, the link-local IPv6 addresses used are randomly generated (to avoid using the device's real MAC address, which could be tracked) ³⁴ . The AirDrop protocol, as implemented in 2017–2020, is fully encrypted end-to-end between the two devices ³⁵ .

- **Data transfer:** The sending device (say, the MacBook Pro) then streams the file data directly over the Wi-Fi link to the receiver (the iMac). Because this uses Wi-Fi (802.11ac) with no internet bottleneck, the transfer speeds can be very high – often on the order of tens of megabytes per second. In ideal cases, AirDrop can send a multi-gigabyte video file in under a minute or two. There is effectively no fixed file size limit (Apple doesn't specify a limit, though extremely large transfers may falter due to timing out or memory usage) ³⁶ . Both of our example Macs, with 802.11ac 3x3 radios, can achieve top AirDrop throughput when talking to each other. Users have reported AirDrop sending, say, a 1 GB file in ~30 seconds under good conditions – vastly outpacing legacy methods like Bluetooth file transfer or copying via USB 2.0 stick.
- **Security and privacy:** AirDrop has a few security layers. First, as noted, all transfers are encrypted (TLS). Second, if the receiving device is set to “Contacts Only” mode, an authentication occurs where the devices exchange hashed identifiers derived from the users' contact info to verify they know each other. Importantly, researchers in 2019 identified that during this handshake, devices send out a **SHA-256 hash** of their owner's phone number and email to each other to check for a match in contacts ³⁷ . While the hashes are not plaintext, a determined attacker could potentially brute-force phone numbers or small email dictionaries to find a match, raising privacy concerns. Apple has since provided warnings about this and in iOS 16.2 introduced a time limit for “Everyone” mode (to mitigate unwanted sharing). But in our 2017–2020 timeframe, users needed to be aware that AirDrop being set to Everyone could allow unknown devices to send prompts, and the aforementioned hash exposure was a theoretical risk if an attacker was sniffing BLE traffic. On the whole, however, AirDrop's security model – requiring acceptance for each file and using encrypted transfer – kept casual usage safe ³⁵ .
- **Protocol evolution:** Initially, AirDrop on Mac (OS X Lion through Mavericks) was a separate implementation incompatible with iOS AirDrop. In 2014, Apple unified the protocol: since OS X Yosemite, Macs use the same AirDrop protocol as iPhones ³⁸ . Thus our iMac and MBP can AirDrop with iPhones (running iOS 8+). There was a “legacy mode” on older Macs to still transfer to pre-2012 Macs over an ad-hoc Wi-Fi (without BLE), but by macOS Mojave in 2018 that legacy mode was removed ³⁸ . Both the 2017 iMac and 2019 MBP, being new machines, simply use the modern BLE+AWDL AirDrop method.

In practical ecosystem workflows, AirDrop became a go-to method for quick sharing: e.g. a user editing photos on the MacBook Pro could AirDrop selected images to the iMac for further editing on a larger screen (or vice-versa). The peer-to-peer link is typically faster than uploading to iCloud Drive and redownloading, and doesn't require any cables or intermediate devices. This emphasizes Apple's strategy of making local connectivity effortless and fast, encouraging users with multiple Apple devices to use them synergistically.

Another peer-to-peer feature is **AirPlay** (Mirroring and direct streaming). In our timeframe, a Mac could AirPlay its screen or media to an Apple TV or AirPlay-enabled speaker. This usually runs over a Wi-Fi network (using Bonjour for discovery and then RTSP streaming). However, Apple TVs from 2015 onward support *peer-to-peer AirPlay*, which again uses BLE for discovery and AWDL for streaming if no common Wi-Fi is

present. For instance, a MacBook Pro brought into a conference room could find a nearby Apple TV (not on any network) via Bluetooth and then stream its display directly to it over an AWDL link. The underlying technology is similar to AirDrop's – creating an ad-hoc secure wireless channel. These peer-to-peer AirPlay sessions benefit from 802.11ac speeds for high-resolution mirroring with minimal lag. The 2019 MacBook Pro, with its powerful CPU/GPU, can smoothly AirPlay mirror its Retina display (scaled to 1080p or 4K on an Apple TV 4K) in real-time. The 2017 iMac can likewise mirror but being less portable, it's more often on Ethernet or Wi-Fi and would use infrastructure AirPlay. Still, the capability is there – reflecting Apple's emphasis that any two devices should communicate directly when needed.

In summary, AirDrop exemplifies the tight integration of connectivity layers: Bluetooth for discovery/authentication and Wi-Fi (via AWDL) for high-throughput data. Both the iMac and MacBook Pro fully support these features, enabling workflows like dropping files, clipboard content, or web URLs to each other instantly. The combination of Thunderbolt for peripherals and AirDrop/AirPlay for devices and data means our example Macs are extremely well-connected in both wired and wireless domains.

Apple Continuity and Inter-Device Integration

Beyond raw network connections, Apple's ecosystem is distinguished by **Continuity** – a suite of features that allow multiple Apple devices to act as an integrated workflow. Between 2017 and 2020, Apple expanded Continuity with new capabilities like Universal Clipboard (2016), Auto Unlock with Apple Watch (2016), and Sidecar (2019). These features typically require a mix of Bluetooth, Wi-Fi, and iCloud connectivity to function. They are designed to be seamless for the user, leveraging whatever connection is available (often BLE and AWDL behind the scenes) to link devices. Our 2017 iMac and 2019 MacBook Pro, running modern macOS (e.g. Ventura in 2023, which they both support), benefit from the full Continuity feature set available in that era.

Some key Continuity features and how they work:

- **Handoff:** Arguably the cornerstone of Continuity, Handoff lets you start a task on one device and continue on another. For instance, you could start writing an email on the MacBook Pro, then pick up your iPhone and swipe up to instantly continue composing that email in iOS Mail. Technically, when you have a Handoff-capable app open (Mail, Safari, Pages, etc.), the device advertises a Handoff BLE signal containing an encrypted "activity" token. Another device logged into the same iCloud account picks up the signal and queries iCloud to authenticate and obtain the handoff data (like the URL or document state) ³⁹. The transfer of context might occur via an iCloud handshake or directly over Wi-Fi if both are on the same network. Both example Macs support Handoff (all Macs 2012 and newer do) ⁴⁰. So the iMac 5K and MBP can handoff documents or web browsing sessions to each other or to an iPhone/iPad. They use Bluetooth to know they're near each other and internet (or peer Wi-Fi) to send the actual data. Handoff was extended in 2020 to include FaceTime calls (continuity of calls) and even media playing (Handoff music from iPhone to HomePod, etc.), but during 2017–2019 it mostly covered app activities. The presence of Bluetooth 4.2 vs 5.0 isn't really noticeable for Handoff – both work as long as devices are in range.
- **Universal Clipboard:** Introduced with macOS Sierra and iOS 10, this feature allows you to copy on one device and paste on another. For example, on the iMac you copy a snippet of text or an image, and then on the MacBook Pro you can simply hit paste and the content appears. Universal Clipboard transfers the clipboard data securely over Bluetooth/Wi-Fi between devices logged into the same

iCloud (using an end-to-end encrypted channel). It requires devices be near each other and have Handoff enabled ³⁹. Both the 2017 iMac and 2019 MBP support this (requirement is Sierra or later on a 2012+ Mac) and it greatly facilitates multi-device workflows – e.g. copy a screenshot or a code snippet from the MBP and paste into a document on the iMac ³⁹. The latency is usually under a second or two, thanks to BLE triggers and fast Wi-Fi exchange. This showcases Apple's ability to abstract the networking – the user just experiences a unified clipboard across devices.

- **Auto Unlock with Apple Watch:** If the user has an Apple Watch, their Mac (iMac or MacBook) can automatically unlock when the Watch is nearby and authenticated. This feature, added in 2016, uses Bluetooth proximity to detect the watch and then a Wi-Fi direct connection to quickly exchange unlock tokens. The 2019 MacBook Pro, with its T2 chip and Touch ID, also supports Watch unlock (the presence of Touch ID doesn't preclude it). The 2017 iMac (no Touch ID) relies on Auto Unlock or typing a password. Both models support Auto Unlock as long as they have Bluetooth LE (which they do) and are running macOS 10.13+ with an Apple Watch running watchOS 3+ ³⁹. This is an example of inter-device integration boosting convenience, and it's secured by using the watch's authentication (the watch must be on the user's wrist, unlocked, and recently authenticated by passcode or unlock of iPhone). The Mac senses the watch's presence at wake via BLE and then uses a Wi-Fi session (or internet) to validate credentials with Apple's servers, finally unlocking the Mac almost instantly.
- **Phone and Messaging Continuity:** With an iPhone on the same network, the iMac and MacBook Pro can make and receive cellular calls (routing through the iPhone) and send/receive SMS texts (via the Text Message Forwarding feature). This works via Wi-Fi or wired network: the Mac and iPhone must be on the same LAN (or use AWDL if no router). When a call comes to the iPhone, it uses Bonjour over Wi-Fi to alert the Mac, which then uses the iPhone as a VOIP relay. Both example Macs fully support this (requirement: iPhone on iOS 8.1+ and Mac on Yosemite+ on same network) ⁴¹. It effectively turns the Mac into a speakerphone for the iPhone. Similarly, an SMS sent to your phone can appear in the Mac's Messages app and you can reply from the Mac – the iPhone will forward the SMS. These features cement the multi-device workflow: one can stay on the Mac for communication instead of constantly picking up the phone.
- **Sidecar:** A major addition in macOS Catalina (2019) was **Sidecar**, which allows a modern iPad to act as a secondary display (and drawing tablet) for a Mac. The 2019 MacBook Pro and 2017 iMac both meet the supported hardware list for Sidecar (which requires relatively recent models and OS Catalina+) ⁴². Sidecar can work wired (via USB cable) or wirelessly. Wirelessly, it again leverages a direct point-to-point Wi-Fi connection (likely AWDL or a form of Wi-Fi Direct) between Mac and iPad, with Bluetooth used to initiate/handshake ⁴³. This is essentially Apple's answer to products like Duet Display but built-in and optimized. The MacBook Pro 16", being quite powerful, can drive the iPad at 60 fps as an external display with minimal lag, and supports the Apple Pencil input from the iPad as an input to Mac apps. The iMac 2017, while slightly older, also supports Sidecar (Apple's criteria included iMac late-2015 5K or newer) ⁴². Using Sidecar, an artist could use the iPad Pro as a drawing tablet for the iMac, or a user could have an iPad as a portable second monitor for the MacBook Pro when traveling. It's a prime example of Apple-specific inter-device integration leveraging wireless connectivity – effectively streaming an interactive display over Wi-Fi.
- **Other Continuity Features: Instant Hotspot** allows the Mac to remotely activate an iPhone's hotspot and connect to it without entering a password ⁴⁴ ⁴⁵. This uses iCloud authentication and

BLE to signal the iPhone. Both Macs support this, so if no Wi-Fi around, the iMac could automatically tether via a nearby iPhone's LTE, for example. **Continuity Camera** (introduced in 2018) lets you use an iPhone to capture a photo or scan a document and have it appear instantly in a document on the Mac. Under the hood, the Mac sends a request (via BLE or network) to the iPhone's camera app and transfers the image over Wi-Fi. The iMac and MBP on macOS Mojave+ can utilize this to, say, scan a paper using iPhone's camera and insert it into a Pages document on the Mac – another cross-device workflow booster. **Apple Pay Handoff:** The MacBook Pro 16" has Touch ID, which can be used for Apple Pay transactions on the web. The 2017 iMac lacks any biometric sensor, but Apple allows those users to confirm Apple Pay purchases by using an iPhone or Apple Watch as the authenticator (it pops up a prompt on phone/watch). That is yet another integration point: the Mac and iPhone securely communicate (via Bluetooth and internet) to authorize a payment, with the phone's Touch ID or watch's skin contact serving as proof of identity.

Overall, these Continuity features are enabled by a combination of technologies: iCloud (to tie devices to one user and sync some data), Bluetooth LE (for proximity and triggers), Wi-Fi (for bulk data or direct connections), and increasingly the T2/secure enclave for authentication. Both the 2017 iMac and 2019 MacBook Pro are *peers* in this ecosystem – meaning neither is left out. They can interchange roles freely (e.g., either one can be the source or target of a Handoff or AirDrop). The main difference is the MacBook Pro, with a T2 chip, supports "Hey Siri" hands-free (the T2's always-on audio processor listens for the Siri trigger). The iMac 2017, lacking T2, did not initially support always-listening Siri until macOS Catalina allowed using "Hey Siri" if an iPhone/HomePod was nearby. Also, the MacBook's Touch Bar (if relevant) could be considered an integration with Touch ID – but that's internal to the device. From an ecosystem perspective, what stands out is how these Macs can act as extensions of the user's other Apple devices. For instance, one could start editing a document on the MacBook Pro on the go, then come home and use Handoff or iCloud Drive to seamlessly continue on the iMac's big screen and keyboard. They could copy text from a website on the iMac and paste it into a note on the MacBook. If the user gets a phone call, all their devices ring and they can pick it up on the MacBook's microphone. Such continuity makes the sum of owning multiple Apple products greater than the parts – a strategic advantage Apple has cultivated. Technically, the tight synchronization relies on those connectivity advances (fast Wi-Fi, BLE broadcasts, etc.) and sophisticated software coordination (e.g., the Continuity frameworks introduced in iOS 8/OS X Yosemite).

System Architecture, Security, and Deployment Considerations

During 2017–2020, Apple made significant changes under-the-hood that affected how Macs interface with operating systems and peripherals. A major development was the introduction of the **Apple T2 Security Chip** (Apple's custom ARM co-processor) in Mac models. The 2019 16" MacBook Pro features a T2 chip (as do all 2018–2019 MacBook Pros), whereas the 2017 iMac does not have a T2 (the T2 debuted in the iMac Pro later in 2017, but the standard iMac retained the traditional architecture). This difference has implications for security, I/O processing, and deployment workflows:

T2 Chip and I/O: The T2 in the MacBook Pro acts as a controller for several hardware functions: it's essentially an embedded computer running a variant of iOS ("bridgeOS") that handles the SSD controller, Touch ID sensor, FaceTime HD camera processing, audio codec, and the Secure Enclave for cryptographic operations. From a connectivity standpoint, one notable aspect is that the T2 presents the internal SSD as an encrypted APFS volume to the main Intel CPU – providing on-the-fly encryption/decryption with almost no performance loss. It also means that removing the SSD (which on T2 laptops is actually soldered, not

removable) would not let you read the data without the T2. The T2 also can **block certain ports or boot options for security**. For example, by default the T2's **Startup Security Utility** disallows booting from external media ⁴⁶. This is a stark change from earlier Macs: on the 2017 iMac, one can plug in an external USB or Thunderbolt drive and boot from it freely (or netboot from a network server) as long as you have the right boot selection. On the MBP 2019, unless the user goes into Recovery and lowers the security settings, the T2 will refuse to boot from an external disk or network ⁴⁶. This affects workflows like macOS installation and troubleshooting – for instance, deploying a lab of 2019 MacBook Pros via NetBoot/Imaging is not possible by default, whereas the 2017 iMac can still use older NetInstall/DeployStudio workflows ⁴⁶. Apple's intention with T2 was to enforce **Secure Boot**: by default the T2 in Full Security mode will only boot a legit, signed copy of macOS and only from the internal disk (which it cryptographically seals) ⁴⁷ ⁴⁸. This improves security (no Evil Maid attacks via USB stick) but did “kill off” the traditional monolithic imaging deployment method in IT departments ⁴⁶. Instead, institutions had to transition to Apple's Device Enrollment Program (DEP) or use Internet Recovery installation. The 2017 iMac, lacking T2, can still be imaged or booted from external media without such hurdles.

Deployment & Target Disk Mode: Another related feature is **Target Disk Mode**, historically used to connect a Mac as an external drive to another Mac (via FireWire or Thunderbolt). The iMac 2017 supports Thunderbolt Target Disk Mode – you could connect it to, say, the MacBook Pro with a TB3 cable, restart the iMac in Target Disk Mode, and it would appear as an external volume on the MBP for data transfer or even booting. On T2 Macs, Target Disk Mode behavior changed into “**Share Disk**” in the macOS Recovery, where the T2 Mac can expose its internal drive over USB or Thunderbolt but only after authenticating in Recovery. The MBP 16 can do a form of target disk mode, but because of Secure Enclave tied to the disk, you might need to input the password before it will share. Apple also introduced a special DFU mode on T2: if the T2 firmware is corrupted, the MacBook Pro can be plugged into another Mac running Apple Configurator to revive the T2. The iMac 2017 doesn't have this complexity – it has a standard EFI firmware that can be revived through a normal USB restore if needed, and no second computer/Configurator workflow is necessary. From an IT support viewpoint, the iMac was simpler to recover (Internet Recovery or netboot imaging), whereas the T2-equipped MacBook required new procedures (internet recovery with T2 verification, etc.). Indeed, by 2019 Apple was steering customers toward using **Internet Recovery and MDM** to deploy Macs. Imaging an OS onto the MBP 2019 would require first booting to Recovery and disabling secure boot or allowing external boot, which is an extra manual step ⁴⁶. In contrast, one could network-boot 20 iMacs and blast an image to them (though Apple didn't “support” it officially anymore, it was still technically possible).

Security Benefits: The T2 chip brings strong security benefits to the MacBook Pro. It has a Secure Enclave that stores Touch ID data and encryption keys. FileVault disk encryption is automatic and hardware-accelerated. The T2 also enforces that the OS hasn't been tampered with (it will only boot a macOS signed by Apple – in fact, in Full Security, the T2 will *personalize* the macOS installer to that specific Mac using the unique chip ID ⁴⁹ ⁵⁰). This means the MBP's system integrity at boot time is extremely high. By contrast, the iMac 2017 relies on the older firmware password and EFI Secure Boot which was much less rigorous (Apple did not implement UEFI Secure Boot on most Intel Macs except via the T2). The iMac can boot unsigned OSes (Linux, Windows via BootCamp, etc.) freely. The MBP 2019 *can* boot Windows via Boot Camp, but even Windows had to be signed by Microsoft in a way Apple's T2 would accept (and Boot Camp Assistant configures the T2 to allow it) ⁴⁷. Booting Linux or other OS on a T2 Mac required setting the T2 to Reduced Security or No Security mode in the utility ⁵¹. So Apple drew a line in the sand: post-2018 Macs would be much more appliance-like and secure by default, whereas 2017 and prior were more open but less protected. This is something enterprises had to adapt to – e.g., **Firmware updates:** on the iMac, firmware

updates came as part of macOS updates and could be applied via imaging if needed. On the T2 MBP, the firmware is on the T2 and gets updated with OS installs; also, if the T2 is wiped, one might need to use Apple Configurator to revive. These details underscore how the connectivity of internal components changed (the SSD is on the T2's PCIe lanes, etc.).

Performance and OS Support: In terms of raw performance, the connectivity differences (TB3 speeds, Wi-Fi bandwidth) mean both Macs are well-equipped for the demands of macOS in that era. The iMac's 1 TB Fusion Drive, while offering ample storage, is slower than the MBP's 1 TB NVMe SSD. Users dealing with large media might connect an external USB-C SSD to the iMac to improve disk I/O, whereas the MBP's internal SSD is already extremely fast (measured around 2.5–2.8 GB/s read/write ⁵² ⁵³). For multi-device OS workflows, both can run the latest macOS (Ventura in 2023) but notably **macOS 14 Sonoma (2024) dropped support for the 2017 iMac** while retaining the 2019 MBP, partly because Sonoma requires either a T2 or Apple Silicon for certain features. This highlights an “upgrade path” difference: the 2017 iMac is at the end of its OS line (Ventura is the last supported OS for non-Pro 2017 iMacs ⁵⁴ ⁵⁵), whereas the 2019 MBP likely has a couple more years of OS support thanks to the T2 and its newer architecture. This is relevant for deployment: running the latest OS is important for security; the MBP will be able to run macOS 14, the iMac not without unofficial patching. Thus, in the broader ecosystem, the T2 became a sort of dividing line for longevity and capability – Apple clearly optimized features for T2/Apple Silicon moving forward (e.g., **Virtualization:** the T2 MacBook can use Apple's Hypervisor frameworks with hardware assist for VMs, etc., whereas the iMac lacks some of those enhancements).

Legacy Device Integration: The iMac, with more ports, can interface with older devices directly – for example, target disk mode with a FireWire drive (via a Thunderbolt 3 to Thunderbolt 2 adapter then Apple TB2-to-FireWire adapter) is possible. The MBP requires dongles for any non-USB-C device, which can complicate certain connections (e.g., connecting to an older projector via VGA means two adapters daisy-chained from USB-C). For deploying labs, the iMac has that built-in Ethernet which can utilize NetInstall or be imaged en masse by plugging into a switch. The MBP, if an institution wanted to netboot it (for, say, diagnostics), simply *cannot netboot* at all in a traditional sense due to the T2's restrictions ⁵⁶ . They would instead have to use Internet Recovery or bring up macOS and then use management tools. This was a paradigm shift in Mac deployment between 2017 and 2019: **imaging was effectively deprecated** by Apple by the time of the MBP 16" ⁴⁶ . Organizations had to adapt with workflows like creating a bootstrap package and using DEP/MDM for zero-touch setup, rather than clone images.

In everyday use for a consumer, these differences are subtle – the MBP may prompt for Touch ID or Watch to unlock, the iMac always requires a password or Watch. The MBP might be slightly more locked down if trying to boot alternative OS. But both are extremely capable macOS machines and part of the same ecosystem. The iMac's lack of T2 doesn't impede any *consumer* features (Continuity, etc., all work); it's mainly security and future OS implications.

To summarize, in the 2017–2020 window the **iMac 5K (2017) represents the last generation of Mac that used only standard PC architectures** (Intel CPU + legacy EFI + software-based security), while the **MacBook Pro (2019) showcases Apple's transition to integrated security and control chips (T2)**. Connectivity-wise, the T2 doesn't change external I/O speeds (both have TB3 40 Gb/s), but it does intermediate some of those connections (the SSD, camera, etc. go through T2). For IT deployment and multi-device management, the MBP needed new strategies (no netboot, require MDM) whereas the iMac could be managed with older techniques. Apple's ecosystem benefited from T2 in terms of user experience too (e.g., **Hey Siri** voice activation is enabled on the T2 MacBook Pro but not on the iMac 2017). In multi-

device workflows, both computers can serve in roles like local servers, shared devices, or development stations, but the iMac's upgradability (it has SO-DIMM slots for RAM – our example has 48 GB upgraded – and a drive that can be replaced with an SSD if one opens the chassis) gives it a different “lifespan” profile than the MBP (soldered RAM/SSD, non-upgradeable). The iMac can be refreshed by upgrading RAM or using faster external storage; the MBP mostly relies on external expansions for upgrades (eGPU for graphics, TB3 drives for more storage, etc.). This reflects two philosophies in Apple's lineup coexisting in 2017–2019: a semi-upgradable desktop vs. a sealed, but highly optimized, laptop.

Peripheral Compatibility and Expansion

Both the 2017 iMac and 2019 MacBook Pro support a wide array of external peripherals, though the types of peripherals and how they connect differ due to port configurations:

Displays: The iMac 5K has an internal high-resolution display and can additionally drive external monitors via Thunderbolt 3/DisplayPort. As noted, it supports one 5K external or two 4K externals ¹². These would typically be connected using a USB-C to DisplayPort cable or an Apple Thunderbolt 3 to Thunderbolt 2 adapter (for older Thunderbolt displays). Many users took advantage of the iMac's TB3 ports to hook up an **LG UltraFine 5K** display (co-developed by Apple) which connects via a single TB3 cable. The 2019 MacBook Pro, lacking an internal 5K panel but with a very capable GPU, is often seen driving multi-monitor setups. It can output to dual 6K displays – for instance, two Pro Display XDRs at full 6016×3384 resolution ¹⁸ – a capability only the highest-end Intel Macs of that era had. This makes the MBP 16” suitable for advanced workflows like video editing on multiple reference monitors. Both machines support use of **Docks:** a Thunderbolt 3 dock can be attached to provide additional HDMI ports, USB-A ports, etc., which can particularly benefit the MacBook Pro (e.g. a CalDigit TS3 dock can turn one TB3 port into multiple connections). The iMac, already having many ports, might use a dock mainly to add extra display outputs or faster UHS-II SD card readers, etc.

One thing the iMac cannot do is act as an external display for another Mac (“Target Display Mode” was only supported on iMacs up to 2014 and not with Retina 5K iMacs). So you cannot plug the MacBook Pro into the 2017 iMac and use the iMac as a monitor – a feature some earlier iMacs had via Thunderbolt Target Display. Apple removed that once displays went 5K (citing technical limitations). In 2020+, Apple introduced Sidecar and later Universal Control to let an iMac and MacBook share inputs or screens in a way, but that's not a direct monitor connection. So in our scenario, the iMac and MBP are more likely to use Sidecar with an iPad or external monitors than to directly combine with each other's display.

External GPUs: Both Macs, thanks to Thunderbolt 3, support external GPU enclosures (eGPU) using PCIe bandwidth over TB3. Apple's macOS High Sierra and Mojave provided official support for AMD GPU eGPUs. A 2017 iMac user might use an eGPU to supplement the internal Radeon Pro 575/580, for example, adding a Vega 64 in an enclosure to accelerate VR or improve gaming performance. The 2019 MBP with an already powerful Radeon Pro 5500M could still benefit from a desktop-class GPU (like a Radeon 5700XT in eGPU) for tasks like driving multiple 4K game engines or speeding up certain OpenCL/Metal computations. Both machines connect to eGPUs via a TB3 cable; the performance is slightly limited by the 40 Gb/s (~< 4 GB/s after overhead) of Thunderbolt, but still very useful. Apple even sold a Blackmagic eGPU which was commonly paired with MacBook Pros. The iMac being static might be less in need of an eGPU (since it has discrete graphics and one can opt for higher-tier GPU in BTO), but it's still a viable upgrade path, especially as new GPUs came out after 2017. The MacBook Pro, being non-upgradeable internally, relies on eGPU if the user needs more graphics horsepower down the line. This demonstrates how Thunderbolt 3 expands the

upgrade paths for otherwise closed systems like the MBP – you can attach not only GPUs but also high-end audio interfaces, RAID storage, or even specialized devices like PCIe breakout boxes for scientific instrumentation.

Storage and I/O: For external storage, both the iMac and MBP can use USB-C or USB-A drives. The iMac's USB-A ports are USB 3.0 (5 Gb/s), which is fine for hard drives and SATA SSDs (~500 MB/s max). The MacBook's USB-C ports in USB mode support USB 3.1 Gen2 (10 Gb/s) ⁵⁷, so a USB 3.1 Gen2 SSD (NVMe over USB enclosure) can run at up to ~1 GB/s on the MBP. If the iMac user wants similar speed, they could plug the same USB-C SSD into one of the Thunderbolt 3 ports (via USB-C cable) since those support 10 Gb/s USB 3.1 as well ⁵⁸. Thus, both effectively support the latest USB speeds; the difference is the MBP has only USB-C connectors, the iMac has both USB-C and older USB-A 5 Gb/s ports. Compatibility with USB 2.0 devices (like Arduinos, older printers, etc.) is there on both: the iMac's USB-A are backwards compatible ⁵⁹, and the MBP can handle them via USB-C adapter or hub.

One area of note is **Ethernet**: the iMac's built-in Gigabit Ethernet port can be upgraded (on the 2017 iMac, 10 Gigabit Ethernet was not offered – that was on the iMac Pro and 2019 iMac 27 as an option). So the 2017 iMac is limited to 1 Gb/s wired unless using a TB3-to-10GbE adapter. The MacBook Pro has no Ethernet, so users commonly use a Thunderbolt 3 or USB-C Ethernet adapter when needed. The lack of Ethernet on the MBP is a limitation for certain pro workflows (like low-latency audio network streaming or simply environments that require wired connections for security). However, it underscores Apple's trend to remove built-in LAN on portables since 2012. Many creative pros carry a small USB-C dongle that provides an RJ-45 jack when needed.

Audio and Video peripherals: Both Macs have a 3.5 mm audio output (and the iMac's doubles as line-in). They can connect to pro audio interfaces via USB or Thunderbolt easily. For example, a musician might use a Thunderbolt audio interface (like an Apollo) – this can plug into either the iMac or MBP TB3 port. MIDI keyboards or controllers that use USB-A would plug directly into the iMac, but need a USB-C hub/dongle on the MBP. On the video front, for connecting cameras or SD cards: the iMac's SDXC slot (UHS-II capable) is convenient for photographers. The MBP 16" notably lacks an SD slot (Apple brought it back in the 2021 models after feedback) ¹⁶. So an SD card from a camera would require a USB-C SD reader on the MBP, whereas the iMac accepts it directly. This difference is frequently cited by photographers – the 2019 MBP not having SD slot was a pain point.

Backward Compatibility: Through the right adapters, both Macs can interface with older standards: e.g., the iMac's TB3 can adapt to Thunderbolt 2 (for an old Thunderbolt Display or a FireWire RAID via TB2->FW adapter). The MBP can do the same with the official Apple TB3->TB2 adapter (it supports display and storage target devices). Neither machine has optical drives, but one can use a USB external DVD drive on both (works via the iMac's USB-A or the MBP's USB-C-with-dongle). Bluetooth 4.2 vs 5.0 doesn't affect accessory compatibility – a Bluetooth 5.0 device like newer BLE sensors or a BLE stylus will still work with the iMac's BT4.2 (just at BT4 speeds), since BT5 is backward compatible ⁶⁰. So practically, all Bluetooth accessories (mice, keyboards, headsets, etc.) that one would use on the MBP work on the iMac too.

To illustrate the **connectivity and expansion differences** between the two models, the following table compares their key I/O and expansion features:

Feature	iMac 5K (2017)	MacBook Pro 16" (2019)
Thunderbolt 3 Ports	2 × TB3 (USB-C), 40 Gb/s each ⁹	4 × TB3 (USB-C), 40 Gb/s each ¹⁵
USB Ports	4 × USB-A 3.0 (5 Gb/s) ⁵⁹ (plus USB 2.0 compatible)	0 × USB-A (use adapters); USB 3.1 Gen2 (10 Gb/s) support on USB-C ⁵⁷
Display Support	Internal 27" 5K + 1× 5K external (60 Hz) or 2× 4K (60 Hz) ¹²	Up to 2× 6K externals (60 Hz) or 4× 4K (60 Hz) ¹⁸
GPU & eGPU	Radeon Pro 570/575/580 (4–8 GB); supports external GPU via TB3	Radeon Pro 5300M/5500M (4–8 GB); supports eGPU via TB3 (for higher performance)
Network	Gigabit Ethernet (RJ-45) + 802.11ac Wi-Fi ¹¹ ²⁰	802.11ac Wi-Fi only (Gigabit via USB-C adapter) ²¹
Bluetooth Version	4.2 ²⁹	5.0 ⁶¹
SD Card Slot	Yes – SDXC UHS-II ⁶²	No (external USB-C reader required)
Audio I/O	3.5 mm headphone (line out/in) ⁶³ ; built-in mic & stereo speakers	3.5 mm headphone out ⁶⁴ ; 6-speaker system, 3-mic array ⁶⁵
Security Chip	None (legacy SMC only)	Apple T2 Security Chip (Secure Enclave, Touch ID)
Unlock Options	Password, Apple Watch Auto Unlock ³⁹	Password, Touch ID, Apple Watch Auto Unlock
OS Boot Options	Supports external boot, NetBoot/Target Disk Mode (no restrictions)	External boot disallowed by default (configurable) ⁴⁶ ; no NetBoot; Target Disk Mode with constraints
User-Upgradeable	RAM upgradable (4 slots, up to 64 GB); 2.5" SATA drive replaceable (Fusion Drive's HDD) ⁶⁶	No internal upgrades (soldered RAM + SSD); only external expansions via Thunderbolt/USB
Continuity Features	Handoff, Universal Clipboard, AirDrop, etc. (full support) ³⁹	Handoff, Univ. Clipboard, AirDrop, + Hey Siri (T2) ³⁹
Sidecar Support	Yes (macOS Catalina+, iMac 2017+ supported) ⁴²	Yes (Catalina+, all MBP 2016+ supported) ⁴²
Max macOS Version	macOS 13 Ventura (macOS 14 not supported) ⁵⁴ ⁶⁷	macOS 14 Sonoma (supported, and likely beyond) ⁶⁸ ⁶⁹

Table: Comparison of Connectivity and Expansion Features of the 2017 iMac 5K vs. 2019 MacBook Pro.

As the table highlights, the iMac offers a wider variety of built-in ports (catering to older peripherals) and some internal upgradeability, whereas the MacBook Pro focuses on an all-USB-C Thunderbolt approach,

relying on external solutions for any additional needs. In multi-device workflows, both can equally participate in Apple's ecosystem features (AirDrop, Continuity), but the MBP's newer Bluetooth and T2 enable a couple of extra conveniences like voice-triggered Siri and Touch ID for fast user switching or Apple Pay.

Conclusion

The 2017–2020 period was pivotal for Apple's hardware connectivity – blending the old and new. The 2017 iMac 5K embodies a bridge between generations: it introduced the high-bandwidth Thunderbolt 3/USB-C standard to the iMac line while retaining familiar ports like USB-A and SD card, thus servicing both cutting-edge and legacy peripherals in a desktop built for power users. In contrast, the 2019 16-inch MacBook Pro represents Apple's uncompromising vision of the future: a streamlined port selection (four versatile Thunderbolt 3 ports) and a reliance on wireless and adaptors for everything else, underpinned by the T2 security/control chip. Despite these differences, both machines leverage **common connectivity technologies – Thunderbolt 3, 802.11ac Wi-Fi, and Bluetooth LE – to interoperate seamlessly** within the Apple ecosystem.

From a **Thunderbolt and bandwidth** perspective, both the iMac and MacBook Pro deliver 40 Gb/s connectivity that in 2017–2019 was state-of-the-art in the PC industry, enabling workflows with 5K/6K displays, ultra-fast external SSDs, and network speeds rivaling wired servers ² ¹⁹. The analysis showed that Thunderbolt 3's adoption (and USB-C connector) was a unifying step – one cable could handle monitors, data, and charging – especially beneficial for the laptop. The iMac's use of TB3 mainly adds expansion (more displays, faster externals) to a stationary setup, whereas the MBP's use of TB3 is fundamental (it *only* has TB3 for I/O, aside from audio). This highlights how Apple tailored connectivity to device roles: the desktop got a complementary mix, the pro laptop went all-in on a single modern standard.

In **wireless connectivity**, both devices exemplify Apple's commitment to robust Wi-Fi and personal area networking. Sticking with 802.11ac through 2020 ensured solid Gigabit wireless performance for tasks like Time Machine backups to AirPort base stations or streaming to Apple TV. Bluetooth 4.2 vs 5.0 differences aside, both Macs fully support Apple's BLE-driven Continuity framework, which is crucial for multi-device experiences ³⁹. We discussed how AirDrop, as a peer-to-peer service riding on AWDL, enables rapid file sharing – essentially treating ad-hoc Wi-Fi as a quick, on-demand personal network ⁷⁰ ³⁵. In an office scenario, one could AirDrop files from the MacBook Pro to the iMac far faster than using a USB stick, thanks to the direct 5 GHz link. Similarly, Sidecar and Handoff leverage the tight wireless integration between macOS and iOS: our MacBook Pro could extend its display to an iPad Pro for on-the-go dual-screen work, then back at the desk, the iMac could pick up the same project via Handoff and allow the user to continue with a full keyboard and mouse. This fluidity is a product of meticulous integration: the devices share context through iCloud and connect through BLE/Wi-Fi without user friction ³⁹.

One can observe how **Apple's ecosystem ethos – “It just works” – is underpinned by these technical capabilities**. For example, Auto Unlock uses the Watch's presence to unlock the Mac, saving time daily; Universal Clipboard and phone call relay blur the lines between devices, making two Macs (or a Mac and iPhone) feel like one cohesive environment ⁴¹ ⁴⁴. The 2017 iMac and 2019 MBP, despite hardware differences, both serve as first-class citizens in this ecosystem. Indeed, a user in 2020 with both machines could use them interchangeably: iCloud Drive syncs files, AirDrop shuttles one-off data, and Continuity features sync clipboard, messages, and more in real-time. The notable differences (like the MBP's Touch ID)

augment certain workflows – e.g. approving an Apple Pay purchase with a fingerprint on the laptop vs. double-clicking an Apple Watch on the iMac – but do not break the continuity chain.

Finally, in terms of **forward-looking upgrade paths**, these years set the stage for Apple's silicon transition. Thunderbolt 3 was later subsumed into **USB4/Thunderbolt 4**, which the Apple Silicon Macs (late 2020 onward) embraced, bringing minor improvements (guaranteed 40 Gb/s for PCIe, support for more displays on docks, etc.). Both the iMac and MBP can interoperate with newer USB4/TB4 devices since TB3 is cross-compatible, ensuring their connectivity isn't obsolete. However, the iMac's lack of Wi-Fi 6 and the MBP's just-in-time inclusion of Wi-Fi 6 in 2020 (with Apple Silicon) meant that by 2021, new Macs had taken another leap. Still, a user with a 2019 MBP and 2017 iMac in 2020 had a robust setup: multi-gigabit wired and wireless options, a plethora of peripheral choices (legacy and cutting-edge), and an ecosystem that allowed the desktop and laptop to complement each other's roles. The iMac could be the heavy-duty workstation (with more ports, easier cooling for long renders, etc.), and the MacBook Pro the portable powerhouse (with identical compute capability on the go). Through iCloud and Continuity, tasks move with the user – illustrating that connectivity is not just about ports and radios, but also about *software interoperability*.

In conclusion, the Apple hardware ecosystem from 2017 to 2020 was defined by **convergence in connectivity**: convergence of data and display (Thunderbolt 3), convergence of devices (Continuity tying macOS, iOS, watchOS together), and convergence of security and functionality (the T2 chip blending roles of many controllers). The 2017 iMac and 2019 MacBook Pro show these trends in action, each with distinct strengths yet working in tandem. Apple achieved an enviable balance where a user could focus on creative workflow rather than file transfer or compatibility issues – a direct result of the high-bandwidth, low-friction technologies described in this analysis. As Apple moved past 2020 into Apple Silicon Macs, many of these connectivity foundations remained (Thunderbolt 3 evolved to TB4, Wi-Fi 6 became standard, Bluetooth 5.x continued, Continuity expanded further with Universal Control, etc.), proving that the 2017–2020 era successfully set the stage for the next generation of Apple's interconnected ecosystem.

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